MODULE 7, LESSON 2

AI 101 FOR MECHANICAL ENGINEERS

Three useful metrics and their limits

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Lesson Structure

1. Foundations: Residuals and Absolute Error

2. Relative Performance: R^2 Formulation

3. Application and System Limitations

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1

Foundations: Residuals and Absolute Error

Establish the baseline error unit and construct aggregate absolute metrics.

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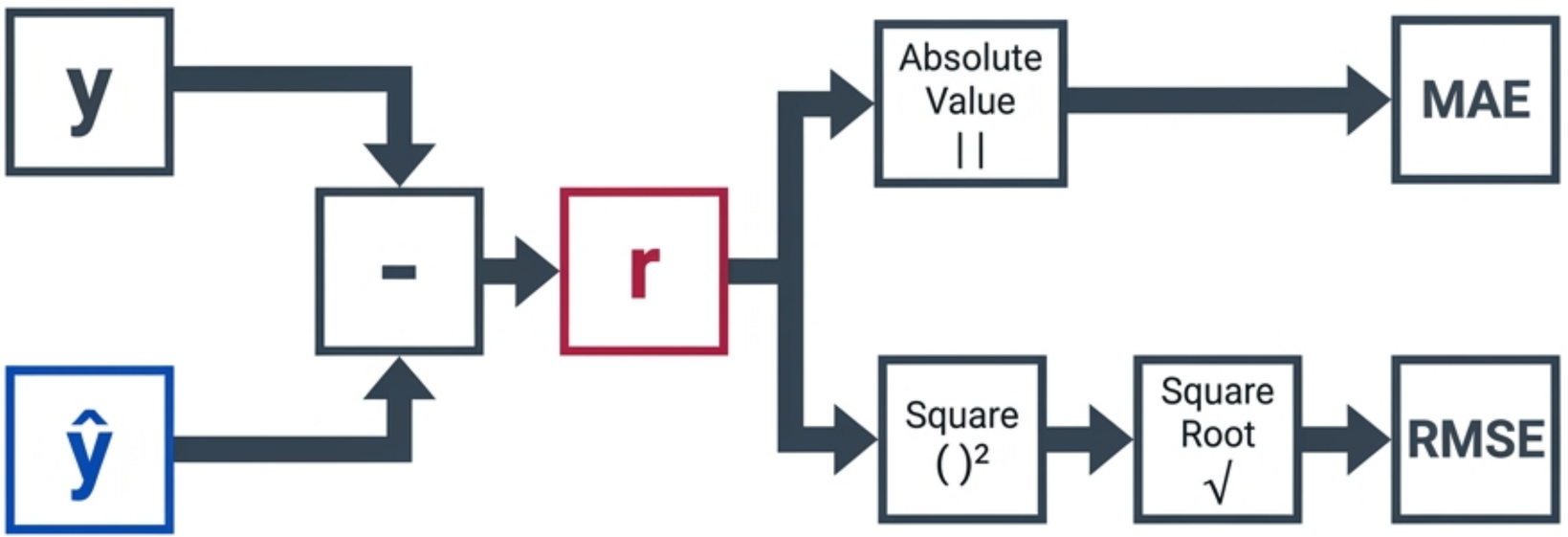
Defining the System Residual

$$r = y - \hat{y}$$

- Observation minus prediction establishes baseline error.
- Variable y denotes measured physical quantity.
- Variable $\hat{y}$ denotes predicted physical quantity.
- Calculations retain target SI units.
- Positive residuals indicate system underprediction.

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MAE and RMSE Computation Flow



- MAE computes the mean absolute residual.
- RMSE computes the root mean square.
- RMSE weights larger system errors heavily.
- RMSE equals MAE only if absolute magnitudes match.

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1. Foundations
2. Relative Performance
3. System Limitations

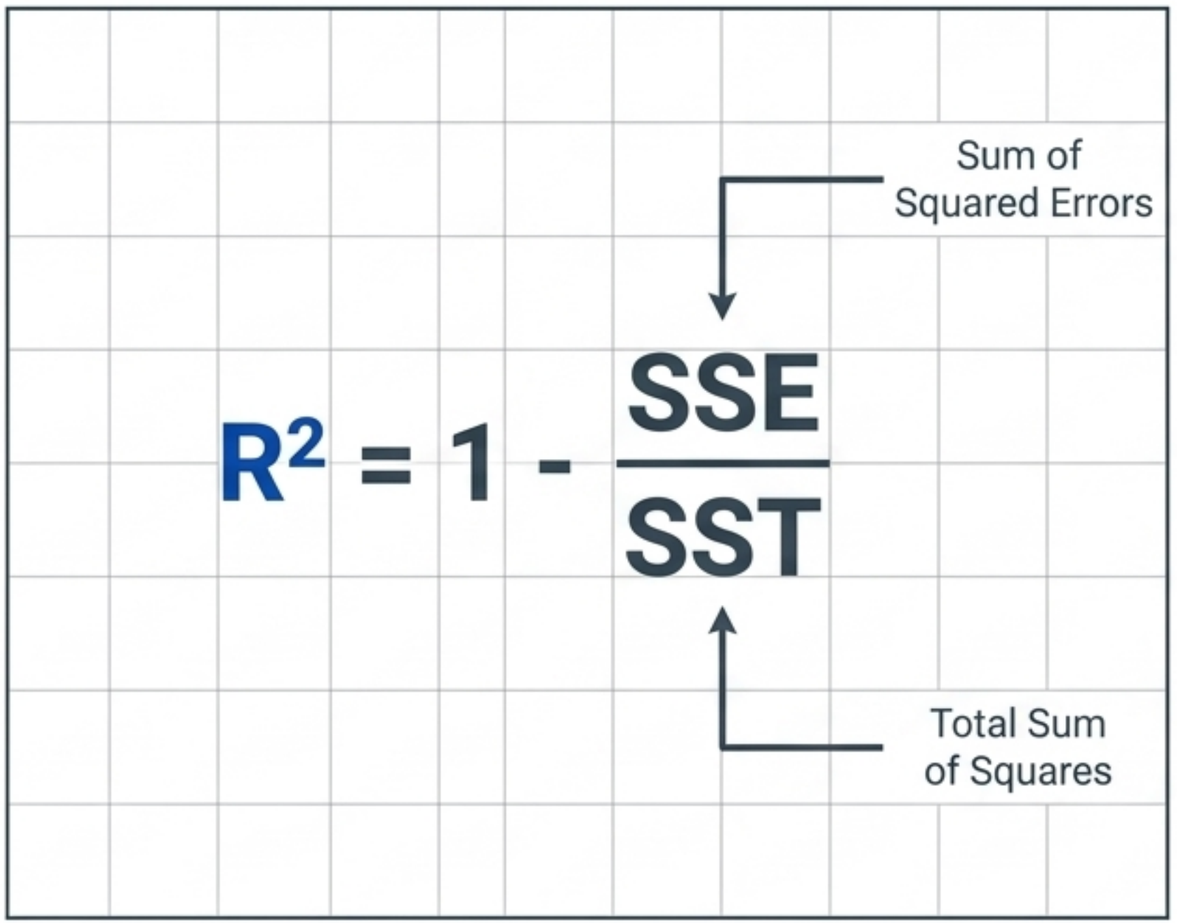
2

Relative Performance: R^2 Formulation

Abstract physical error into a unitless relative statistical score.

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R^2 and the Evaluation Mean



- Compares squared error against evaluation-mean predictions.
- Expressed as unitless relative statistical score.
- Negative values occur if error exceeds baseline.
- Standard denominator becomes zero if target is constant.
- High scores do not establish engineering safety.

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3 Application and System Limitations

Apply metrics to empirical data and analyze systemic diagnostic constraints.

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Worked Calculation: Input Data

Observation	Measured Temp (y in K)	Predicted Temp ($\hat{y}$ in K)
1	5	4
2	8	10
3	10	7

- **Evaluate three** measured temperature rises.
- **Measured** values (**y**) represent the baseline.
- **Predicted** values ($\hat{y}$) represent system outputs.
- **Prepare data matrix** for residual processing.

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Worked Calculation: Outcomes

Observation	Measured Temp (y in K)	Predicted Temp ($\hat{y}$ in K)	Residual (r in K)
1	5	4	+1
2	8	10	-2
3	10	7	+3

MAE = 2 K

RMSE ≈ 2.160 K

R^2 = -0.1053

- Absolute residuals sum to exactly 6 K.
- Mean Absolute Error computes to 2 K.
- Root Mean Squared Error approximates 2.160 K.
- R^2 evaluates to negative 0.1053.
- Negative score dictates immediate residual investigation.

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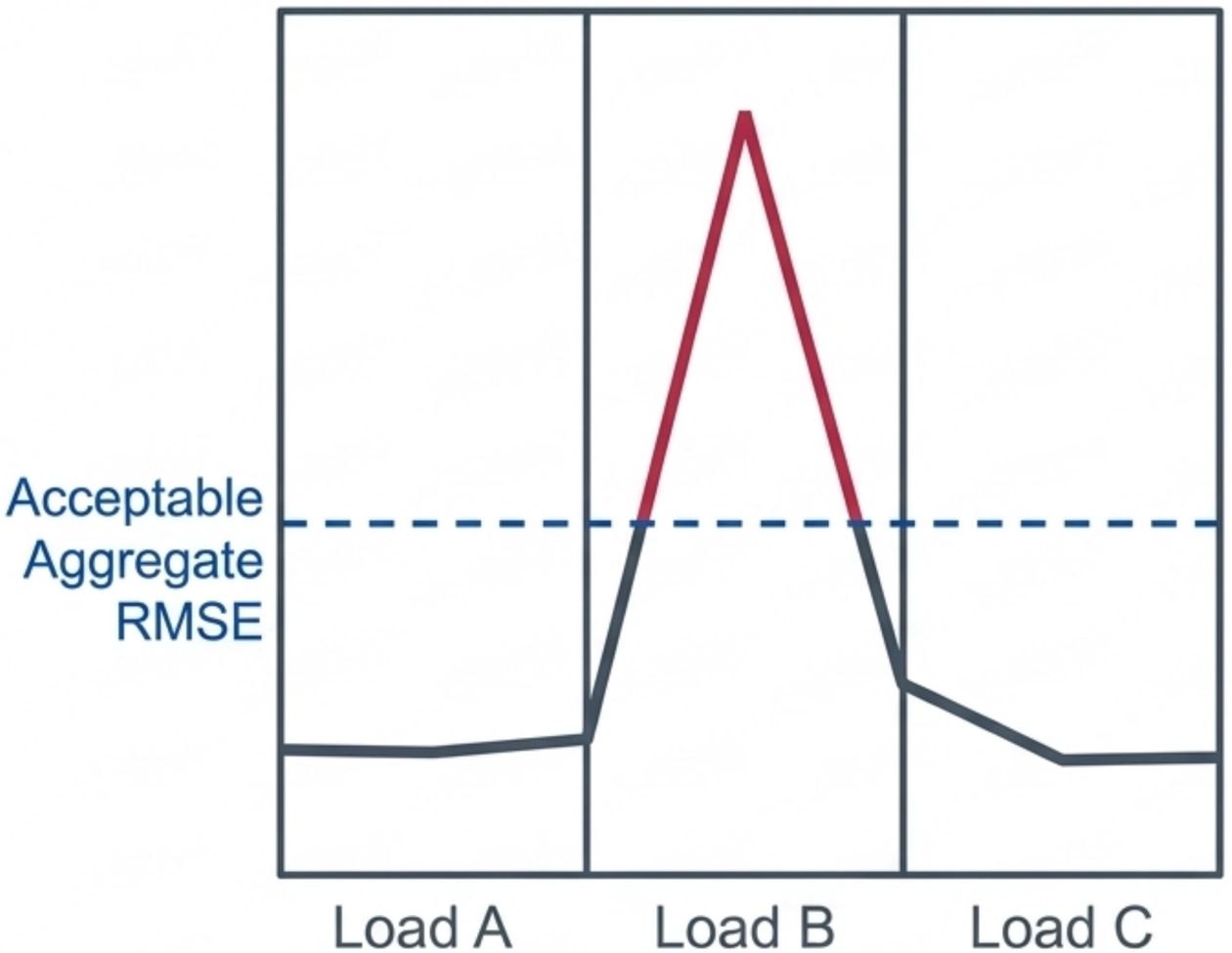
Metric Diagnostic Matrix

Metric	Question Answered	Target Units Present
MAE	Mean absolute residual magnitude	Yes
RMSE	Root mean squared outlier impact	Yes
R^2	Squared error versus evaluation mean	No

- **Matrix** contrasts core **evaluation** metric properties.
- MAE measures average absolute error magnitude.
- RMSE identifies disproportionate outlier impacts.
- **R^2** evaluates relative **variance** without units.
- Always retain sign and units during inspection.

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The Aggregate Error Limitation



- Aggregate metrics obscure narrow severe failures.
- Inspect residuals across distinct operating regimes.
- Verify performance against varying machine loads.
- Analyze error distribution across temporal domains.
- Ensure mathematical scores reflect physical realities.

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Key Points

Foundations: Calculate MAE and RMSE directly from residual values.

Relative Performance: R^2 evaluates variance and can yield negative scores.

Limitations: Aggregate statistical scores routinely hide localized catastrophic system failures.

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